Example 9.59

Find, by integration, the area of the region bounded by the lines $5x-2y=15,\ x+y+4=0$ and the x-axis.

Solution

The lines $5x-2y=15,\ x+y+4=0$ intersect at $(1,-5)$. The line $5x-2y=15$ meets the x-axis at $(3,0)$. The line $x+y+4=0$ meets the x-axis at $(-4,0)$. The required area is shaded in Fig.9.31. It lies below the x-axis. It can be computed either by considering vertical strips or horizontal strips.

When we do by vertical strips, the region has to be divided into two sub-regions by the line $x=1$. Then, we get

$$
\begin{aligned}
A &= \left|\int_{-4}^{1} y\,dx\right| + \left|\int_{1}^{3} y\,dx\right| \\
&= \left|\int_{-4}^{1} (-4-x)\,dx\right| + \left|\int_{1}^{3} \left(\frac{5x-15}{2}\right) dx\right| \\
&= \left|\left(-4x-\frac{x^2}{2}\right)_{-4}^{1}\right| + \left|\left(\frac{5x^2}{4}-\frac{15x}{2}\right)_{1}^{3}\right| \\
&= \left|\left(-\frac{9}{2}\right)-(8)\right| + \left|\left(-\frac{45}{4}\right)-\left(-\frac{25}{4}\right)\right| \\
&= \frac{25}{2}+5 \\
&= \frac{35}{2}.
\end{aligned}
$$

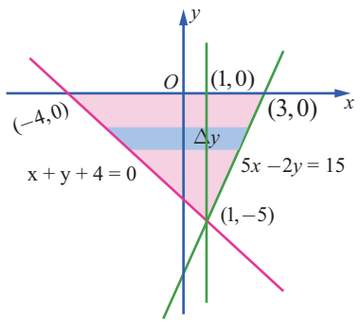


Fig. 9.31

When we do by horizontal strips, there is no need to subdivide the region. In this case, the area is bounded on the right by the line $5x-2y=15$ and on the left by $x+y+4=0$. So, we get

$$
\begin{aligned}
A &= \int_{-5}^{0} [x_R - x_L]\,dy = \int_{-5}^{0} \left[\frac{15+2y}{5} - (-4-y)\right] dy \\
&= \int_{-5}^{0} \left[7+\frac{7y}{5}\right] dy = \left[7y+\frac{7y^2}{10}\right]_{-5}^{0} \\
&= 0-\left[-35+\frac{35}{2}\right] = \frac{35}{2}.
\end{aligned}
$$

Note

The region is triangular with base 7 units and height 5 units. Hence its area is $\frac{35}{2}$ without using integration.

Example 9.60

Using integration find the area of the region bounded by triangle ABC, whose vertices A, B, and C are $(-1,1)$, $(3,2)$, and $(0,5)$ respectively.

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